

Maths

Unit 6 - Fractions
and Percentages.

WALT:

6Nf.02: Understand that a fraction can be represented as a division of the numerator by the denominator.

6Nf.02: Understand that proper and improper fractions can act as operators.

6Nf.07: Recognise percentages of shapes and whole numbers.

WALT: Understand that a fraction can be represented as a division of the numerator by the denominator.
Understand that proper and improper fractions can act as operators.
Recognise percentages of shapes and whole numbers.



**With support, I
can answer
questions.**



**Independently,
I can
answer
questions.**



**I can complete
the “think like
a
mathematician
” exercises.**

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Retrieval Practice

Here are four calculations.

17.2×4

17.09×4

1.72×39

1.7×39

Draw a ring around the calculation that gives the **largest** answer.
You do not need to work out the answers.

[1]

17.2×4

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Exercise 6.1

- 1 Represent these divisions as fractions.
a 5 divided by 6 b 6 divided by 5
c 10 divided by 4 d 4 divided by 20
- 2 Ahmed, Carlos, Ludvik, Oliver and Rajiv share 3 cakes between them. What fraction of a cake does each person get?
- 3 Which word can you use to complete this sentence?
When you find $\frac{6}{5}$ of a number you are using a fraction as an _____.
- 4 Calculate.
a $\frac{3}{4}$ of \$16 b $\frac{5}{4}$ of \$12 c $\frac{5}{2}$ of 4 metres

5 Hal 7 Copy and complete this table to show fractions of 24.

Fraction	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{5}{4}$	$\frac{7}{4}$	$\frac{9}{4}$	$\frac{11}{4}$
Amount	6					

8 Calculate.
a $\frac{9}{4}$ of 16 b $\frac{7}{5}$ of 35 c $\frac{8}{7}$ of 14 d $\frac{4}{3}$ of 15

Think like a mathematician

Imagine that you roll two 1 – 6 dice and use them to make an improper fraction.



$\frac{5}{3}$ is an improper fraction

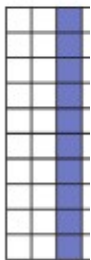
Write all the different improper fractions you could make.

Tip

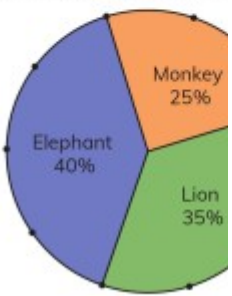
Be systematic.



- 1 Find 10% of these quantities.
a 40 b 70 cm c \$20 d 120 kg
- 2 A man has a mass of 80 kg. 60% of his total mass is water. What is the mass of this water?
- 3 Helga says, 'To calculate 10% of a quantity, you divide it by 10, so to find 50%, you divide by 5.'
Helga is not
- 4 What perce
- 5 What perce
- 6 Calculate.
a 15% of 120 b 45% of 240 c 65% of 180
Check your answers to questions 5 and 6 with your partner.
- 7 Amy has \$400. She spends 40% of her money on a new bike. How much does she spend on the bike?
- 8 Song has strip of paper 20 cm long. He cuts a 5 cm piece off the strip. What percentage of the strip does he cut off?
- 9 a A coat originally costs \$40. The price goes up by 10%. What is the new price?
b A garden centre sells plants at 25% off at the end of the month. The price of a plant was originally \$8. What is its price at the end of the month?



10 The pie chart shows th



32 children chose elepl
How many children are in Stage 6? Explain your answer.

Think like a mathematician

Here are twelve numbers.
10 20 25 40 50 60
75 100 150 160 500 600

Can you write four number sentences using each of the twelve numbers once?

$\frac{\text{blue}}{\text{pink}} \% \text{ of } \frac{\text{green}}{\text{orange}} \text{ is } \frac{\text{red}}{\text{purple}}$

$\frac{\text{light blue}}{\text{dark blue}} \% \text{ of } \frac{\text{light green}}{\text{dark green}} \text{ is } \frac{\text{light red}}{\text{dark red}}$

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Recognise percentages of shapes and whole numbers.

- 1 Represent these divisions as fractions.
- a 5 divided by 6 $\frac{5}{6}$

b 6 divided by 5 $\frac{6}{5}$

c 10 divided by 4 $\frac{10}{4}$

d 4 divided by 10 $\frac{4}{10}$
- 2 Ahmed, Carlos, Ludvik, Oliver and Rajiv share 3 cakes equally between them.
What fraction of a cake does each person get? $\frac{3}{5}$
- 3 Which word can you use to complete this sentence?
When you find $\frac{6}{5}$ of a number you are using a fraction as an operator.
- 4 Calculate.
- a $\frac{3}{4}$ of \$16 **\$12**

b $\frac{5}{4}$ of \$12 **\$15**

c $\frac{5}{2}$ of 4 metres **10 metres**

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- 5 Halima swims $\frac{1}{2}$ of 500 metres and Bella swims $\frac{3}{10}$ of 800 metres.



Halima swims further. One half of 500m is 250 metres. Three tenths of 800 metres is 240 metres.

Who swims further? Explain how you know.

- 6 Leroy and Wayne each have 90 bricks.

Leroy uses $\frac{3}{5}$ of his bricks to build a wall.

Wayne uses $\frac{5}{6}$ of his bricks to build a wall.

How many bricks do they have left altogether? Show your working.

Discuss your answer with your partner. Do you agree?

**Leroy has 90 bricks.
Has $\frac{2}{5}$ remaining = 36**

**Wayne has 90 bricks.
Has $\frac{1}{6}$ remaining = 15**

$36 + 15 = 51$

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7 Copy and complete this table to show fractions of 24.

Fraction	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{5}{4}$	$\frac{7}{4}$	$\frac{9}{4}$	$\frac{11}{4}$
Amount	6	18	30	42	54	66

8 Calculate.

a $\frac{9}{4}$ of 16

36

b $\frac{7}{5}$ of 35

49

c $\frac{8}{7}$ of 14

16

d $\frac{4}{3}$ of 15

20

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9 Write all the numbers from this list that give the result of dividing 23 by 5.

$$4\frac{6}{10}$$

$$4.6$$

$$4\frac{3}{10}$$

$$4\frac{3}{5}$$

$$\frac{23}{5}$$

$$4.3$$

10 Which of these representations are equivalent to $\frac{48}{5}$?

$$5 \div 48$$

$$9.3$$

$$9\frac{3}{5}$$

$$48 \div 5$$

$$9.6$$

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1 Find 10% of these quantities.

a 40 **4**

b 70 cm **7cm**

c \$20 **\$2**

d 120 kg
12 kg

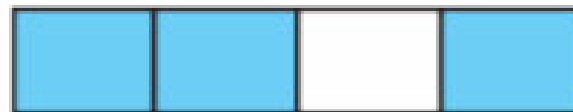
2 A man has a mass of 80 kg. 60% of his total mass is water.
What is the mass of this water? **48 kg**

3 Helga says, 'To calculate 10% of a quantity, you divide it by 10,
so to find 50%, you divide by 5.'

To find 50% you divide by 2

Helga is not correct. Explain what is wrong with her statement.

4 What percentage of the diagram is shaded?

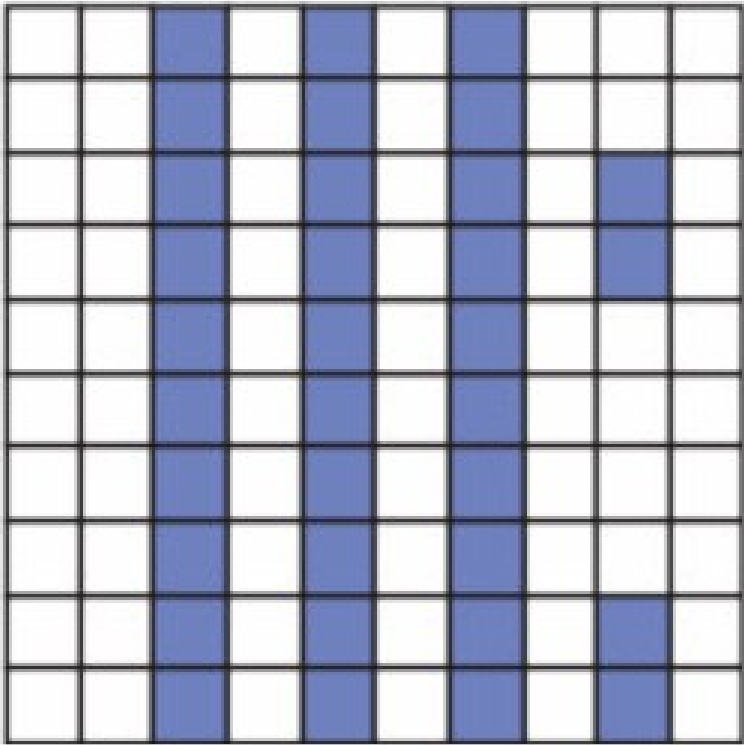


75%

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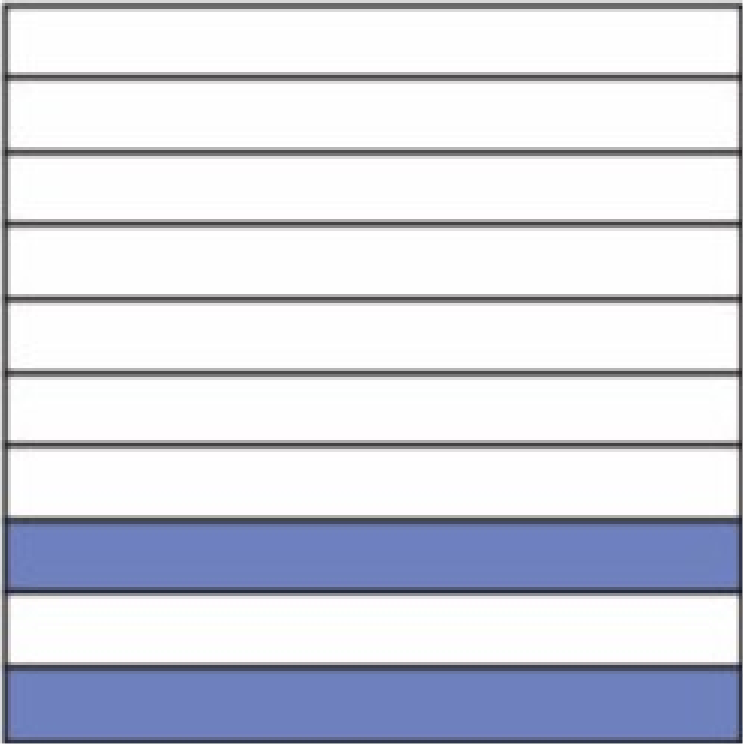
5 What percentage of each diagram is shaded?

a



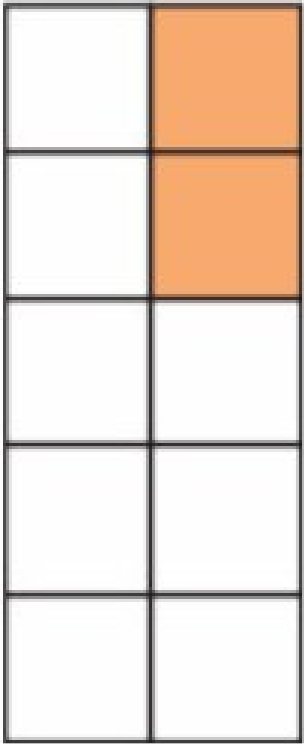
34%

b



20%

c



20%

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6 Calculate.

a 15% of 120 **18** **b** 45% of 240 **108** **c** 65% of 180 **117**

Check your answers to questions 5 and 6 with your partner.

7 Amy has \$400. She spends 40% of her money on a new bike.

How much does she spend on the bike? **\$160**

8 Song has a strip of paper 20 cm long.

He cuts a 5 cm piece off the strip. **25%**

What percentage of the strip does he cut off?



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9 a A coat originally cost \$40. The price goes up by 10%.

What is the new price? **\$44**

b A garden centre sells plants at 25% off at the end of the month.
The price of a plant was originally \$8. What is its price at the end of the month?

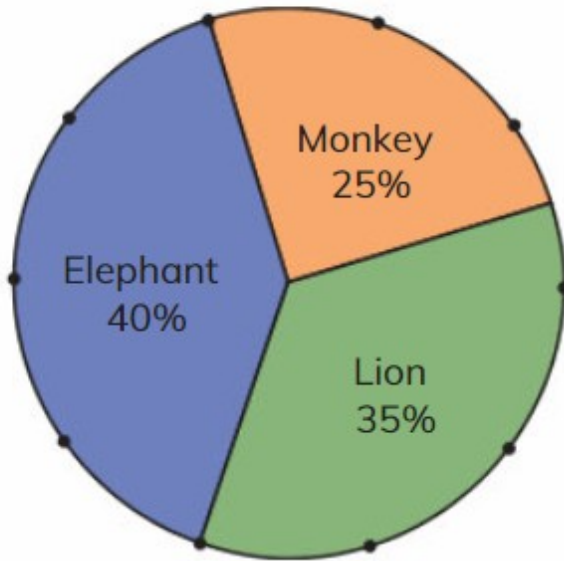
\$6

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10 The pie chart shows the favourite animals of a group of Stage 6 children.



Tip

A pie chart shows a divided circle where each section represents a percentage of the total. In this diagram 40% of the children chose elephants.



32 children chose elephants.

How many children are in the Stage 6 group? Explain your answer.

80 children. We know that 40% is 32 children. If we divide 32 by 4 we get 10% which is 8 children. If 10% is 8 children then 100% must

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